```
!pip install -q ultralytics

from google.colab import drive
drive.mount("/content/drive")
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.9/46.9 kB 1.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.4/1.4 MB 21.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.1/58.1 kB 1.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 97.6/97.6 kB 3.6 MB/s eta 0:00:00
Mounted at /content/drive
```

```
%pip install -q roboflow
%pip install -q roboflow ultralytics
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 373.9/373.9 kB 13.7 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.4/5.4 MB 92.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.4/58.4 kB 6.3 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.9/46.9 kB 2.3 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.4/1.4 MB 44.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 97.6/97.6 kB 12.3 MB/s eta 0:00:00
```

```
from google.colab import userdata
from roboflow import Roboflow

ROBOFLOW_API_KEY = userdata.get("ROBOFLOW_API_KEY").strip()
rf = Roboflow(api_key=ROBOFLOW_API_KEY)
```

```
from roboflow import Roboflow

rf = Roboflow(api_key=ROBOFLOW_API_KEY)

project = rf.workspace(
    "safety-object-detection-app"
).project(
    "construction-site-safety-ccp7o"
)

version = project.version(1)
dataset = version.download("yolov11")

DATA_YAML = f"{dataset.location}/data.yaml"

print("Dataset folder:", dataset.location)
print(open(DATA_YAML).read())
```

```
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in Construction-Site-Safety-1 to yolov11:: 100%|██████████| 140519/140519 [00:03<00:00, 4071

Extracting Dataset Version Zip to Construction-Site-Safety-1 in yolov11:: 100%|██████████| 5206/5206 [00:00<00:00, 6628.30it
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 10
names: ['Hardhat', 'Machinery', 'Mask', 'NO-Hardhat', 'NO-Mask', 'NO-Safety Vest', 'Person', 'Safety Cone', 'Safety Vest', '

roboflow:
  workspace: safety-object-detection-app
  project: construction-site-safety-ccp7o
  version: 1
  license: CC BY 4.0
  url: https://universe.roboflow.com/safety-object-detection-app/construction-site-safety-ccp7o/dataset/1
```

```
from pathlib import Path
import yaml

with open(DATA_YAML, "r") as f:
    config = yaml.safe_load(f)

print("Dataset YAML:", DATA_YAML)
print("Train:", config.get("train"))
print("Validation:", config.get("val"))
print("Test:", config.get("test"))

dataset_dir = Path(DATA_YAML).parent
for split in ("train", "valid", "val", "test"):
    image_dir = dataset_dir / split / "images"
```

```
    if image_dir.exists():
        print(f"{split}: {len(list(image_dir.glob('*')))} images")
```

```
Dataset YAML: /content/Construction-Site-Safety-1/data.yaml
Train: ../train/images
Validation: ../valid/images
Test: ../test/images
train: 1819 images
valid: 518 images
test: 260 images
```

```
from pathlib import Path

DATA_YAML = str(Path(dataset.location) / "data.yaml")
print("Dataset config:", DATA_YAML)
print("Found:", Path(DATA_YAML).is_file())
```

```
Dataset config: /content/Construction-Site-Safety-1/data.yaml
Found: True
```

```
from pathlib import Path
import torch
from ultralytics import YOLO


model = YOLO("yolo11s.pt")
device = 0 if torch.cuda.is_available() else "cpu"

model.train(
    data=DATA_YAML,
    epochs=50,
    imgsz=640,
    batch=16,
    optimizer="SGD",
    lr0=0.01,
    lrf=0.01,
    momentum=0.937,
    weight_decay=0.001,
    workers=8,
    device=0,
    seed=0,
    pretrained=True,
    project="/content/drive/MyDrive/new",
    name="construction_yolo11s_v1",
    plots=True,
)

print("Training output:", model.trainer.save_dir)
```

```
Ultralytics 8.4.179 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module                                       arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv             [3, 32, 3, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  2                  -1  1     26080  ultralytics.nn.modules.block.C3k2            [64, 128, 1, False, 0.25]
  3                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
  4                  -1  1    103360  ultralytics.nn.modules.block.C3k2            [128, 256, 1, False, 0.25]
  5                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
  6                  -1  1    346112  ultralytics.nn.modules.block.C3k2            [256, 256, 1, True]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv             [256, 512, 3, 2]
  8                  -1  1   1380352  ultralytics.nn.modules.block.C3k2            [512, 512, 1, True]
  9                  -1  1    656896  ultralytics.nn.modules.block.SPPF            [512, 512, 5]
 10                  -1  1    990976  ultralytics.nn.modules.block.C2PSA           [512, 512, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 13                  -1  1    443776  ultralytics.nn.modules.block.C3k2            [768, 256, 1, False]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 16                  -1  1    127680  ultralytics.nn.modules.block.C3k2            [512, 128, 1, False]
 17                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 19                  -1  1    345472  ultralytics.nn.modules.block.C3k2            [384, 256, 1, False]
 20                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 22                  -1  1   1511424  ultralytics.nn.modules.block.C3k2            [768, 512, 1, True]
 23        [16, 19, 22]  1    823278  ultralytics.nn.modules.head.Detect           [10, 16, None, [128, 256, 512]]
YOLO11s summary: 181 layers, 9,431,662 parameters, 9,431,646 gradients, 21.7 GFLOPs

Remapped 1/10 cls head rows from pretrained weights by class name
Transferred 499/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
```

```
AMP: checks passed ✅
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1391.6±689.0 MB/s, size: 59.0 KB)
train: Scanning /content/Construction-Site-Safety-1/train/labels.cache... 1819 images, 0 backgrounds, 0 corrupt: 100% ━━━━
train: /content/Construction-Site-Safety-1/train/images/construction-855-_jpg.rf.c1ea9be1efdfc89dd74f1df804dad704.jpg: 1 d
train: /content/Construction-Site-Safety-1/train/images/mask-wearing-1632932085584_png_jpg.rf.8cc368fb53bf958600dcf5b9379a
train: /content/Construction-Site-Safety-1/train/images/n457047_jpg.rf.415d4d962abee66ff2858d9ab724bb38.jpg: 1 duplicate l
train: /content/Construction-Site-Safety-1/train/images/youtube-42_jpg.rf.7ecad9e921e0022f4148062f1f91f305.jpg: 1 duplicat
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_av
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1937.0±619.5 MB/s, size: 48.6 KB)
val: Scanning /content/Construction-Site-Safety-1/valid/labels.cache... 518 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━
optimizer: SGD(lr=0.01, momentum=0.937) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.001), 87 bias(decay=
Plotting labels to /content/drive/MyDrive/new/construction_yolo11s_v1/labels.jpg...
Using 1819 train, 518 val images for fraction=1.0 at imgsz=640
Using 2 dataloader workers
Logging results to /content/drive/MyDrive/new/construction_yolo11s_v1
Starting training for 50 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/50      4.23G      1.409      2.201      1.511        198        640: 100% ━━━━━━━━━━━━ 114/114 3.9it/s 29.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 17/17 2.6it/s
                   all        518       7997      0.539      0.406      0.409      0.229
```

```
from pathlib import Path

runs_dir = Path("/content/drive/MyDrive/new")
run_dir = runs_dir / "construction_yolo11s_v1"

print("Runs folder exists:", runs_dir.exists())
print("YOLO11 run exists:", run_dir.exists())

if runs_dir.exists():
    print("Folders in runs:")
    for item in runs_dir.iterdir():
        print(item.name)

print("Best checkpoint exists:", (run_dir / "weights" / "best.pt").exists())
```

```
Runs folder exists: True
YOLO11 run exists: True
Folders in runs:
construction_yolo11s_v1
Best checkpoint exists: True
```

```
best_model = YOLO(
    "/content/drive/MyDrive/new/construction_yolo11s_v1/weights/best.pt"
)

metrics = best_model.val(
    data=DATA_YAML,
    split="test",
    imgsz=640,
    batch=16,
    device=device,
    plots=True
)

print("Precision:", metrics.box.mp)
print("Recall:", metrics.box.mr)
print("mAP@0.5:", metrics.box.map50)
print("mAP@0.5:0.95:", metrics.box.map)
```

```
Ultralytics 8.4.179 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
YOLO11s summary (fused): 100 layers, 9,416,670 parameters, 0 gradients, 21.4 GFLOPs
WARNING ⚠️ val: Slow image access detected (ping: 0.0±0.0 ms, read: 22.5±10.6 MB/s, size: 62.8 KB). Use local storage inste
val: Scanning /content/Construction-Site-Safety-1/test/labels... 260 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 260
val: New cache created: /content/Construction-Site-Safety-1/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 17/17 2.4it/s 7.
                   all        260       3733      0.889      0.737      0.805      0.567
               Hardhat        131        310      0.887       0.59      0.669      0.444
             Machinery        206        511      0.931      0.869      0.924      0.788
                  Mask        105        164       0.92      0.909      0.934      0.708
            NO-Hardhat        125        223      0.884      0.786      0.833      0.559
               NO-Mask        144        270      0.834       0.67      0.726      0.419
       NO-Safety Vest        185        413      0.885      0.783      0.859      0.597
                Person        252        968       0.91      0.829      0.901      0.681
          Safety Cone         67        455      0.879      0.521      0.626      0.336
          Safety Vest        124        266      0.873      0.677      0.766      0.523
              Vehicle         74        153      0.883      0.742      0.812      0.614
Speed: 2.1ms preprocess, 11.4ms inference, 0.0ms loss, 2.7ms postprocess per image
Results saved to /content/runs/detect/val
Precision: 0.8886822309094283
Recall: 0.7374836293626218
mAP@0.5: 0.8050704361655819
mAP@0.5:0.95: 0.5669558107757983
```

```
from google.colab import drive
drive.mount("/content/gdrive")
```

```
Drive already mounted at /content/gdrive; to attempt to forcibly remount, call drive.mount("/content/gdrive", force_remount=
```

```
from pathlib import Path
import shutil

candidates = [
    Path("/content/drive/MyDrive/new/construction_yolo11s_v1"),
    Path("/content/drive/MyDrive/ppe_detection/runs/construction_yolo11s_v1"),
]

src = next(
    (p for p in candidates if (p / "weights" / "best.pt").is_file()),
    None
)

if src is None:
    print("Could not find best.pt. Check the Colab Files pane for the run folder's actual path.")
else:
    dst = Path("/content/gdrive/MyDrive/new/construction_yolo11s_v1")
    dst.parent.mkdir(parents=True, exist_ok=True)
    shutil.copytree(src, dst, dirs_exist_ok=True)
    print("Copied from:", src)
    print("Copied to:", dst)
    print("Checkpoint copied:", (dst / "weights" / "best.pt").is_file())
```

```
Copied from: /content/drive/MyDrive/new/construction_yolo11s_v1
Copied to: /content/gdrive/MyDrive/new/construction_yolo11s_v1
Checkpoint copied: True
```